

Physics World Models for Computational Imaging: A Universal Physics-Information Law for Recoverability, Carrier Noise, and Operator Mismatch

Chengshuai Yang
NextGen PlatformAI C Corp

Abstract

Computational imaging systems routinely fail in practice because the assumed forward model diverges from the true physics, yet no existing framework systematically diagnoses *why* reconstruction degrades. We introduce Physics World Models (PWM), a universal diagnostic and correction framework grounded in the TRIAD LAW: every imaging failure decomposes into exactly three root causes—recoverability loss (**Gate 1**), carrier-noise budget violation (**Gate 2**), and operator mismatch (**Gate 3**). PWM compiles 64 modalities spanning five physical carriers (photons, electrons, spins, acoustic waves, and particles) into a unified OPERATORGRAPH intermediate representation comprising 89 validated operator templates. Autonomous, deterministic agents diagnose the dominant failure gate and correct the forward model without retraining any reconstruction algorithm. Across 16 validated modalities, correction yields improvements ranging from +0.8 dB to +48.25 dB, with a median gain of +14.5 dB. **Gate 3** is identified as the dominant bottleneck in 14 of 16 modalities, demonstrating that a decade of solver-centric progress has overlooked the principal source of imaging failure. The TRIAD LAW provides the first universal, quantitative language for imaging diagnosis.

1 Introduction

Why do state-of-the-art reconstruction algorithms fail in practice? The answer is deceptively simple: the assumed forward model is wrong, and nobody measures this systematically. The computational imaging community has devoted extraordinary effort to designing ever more powerful solvers—from compressed sensing and plug-and-play priors to end-to-end deep unrolling networks—while treating the forward model as a fixed, trusted input. This implicit assumption is rarely justified. Optical masks shift during assembly, MRI coil sensitivities drift with patient positioning, and CT geometries deviate from their nominal calibration. When these mismatches arise, even the most sophisticated reconstruction algorithms collapse, and the resulting artifacts are routinely misattributed to solver limitations rather than to their true cause: an incorrect physics model.

32 The scale of this crisis is striking. Consider coded aperture snapshot spectral imaging
 33 (CASSI), a representative photon-domain modality. Under ideal conditions—where the
 34 true coded mask is known exactly—the state-of-the-art transformer solver MST-L achieves
 35 34.81 dB on a standard benchmark. Introduce a realistic perturbation of just one pixel of
 36 lateral mask shift, and MST-L plummets to 18.09 dB, a catastrophic loss of 16.72 dB. To
 37 put this in perspective, the cumulative improvement from a decade of solver development
 38 in CASSI—progressing from early iterative methods through deep unrolling to modern
 39 transformer architectures—amounts to roughly 2–3 dB. A single pixel of undiagnosed mask
 40 shift erases more than five times the gains of an entire research generation. This is not
 41 a pathological edge case; analogous degradations appear across modalities, from lensless
 42 imaging to magnetic resonance imaging to computed tomography.

43 The root problem is a missing diagnostic layer. When a reconstruction fails, the prac-
 44 titioner faces a differential diagnosis with at least three distinct failure modes. First, the
 45 measurement may be fundamentally information-deficient: the null space of the forward
 46 operator may preclude recovery regardless of the solver or signal-to-noise ratio. Second,
 47 the carrier budget may be insufficient: too few photons, too low a dose, or too short an
 48 acquisition may push the measurement below the quantum or thermal noise floor. Third,
 49 the assumed forward model may diverge from the true physics: the operator used for recon-
 50 struction may not match the operator that generated the data. These three failure modes
 51 interact, compound, and masquerade as one another, yet no existing framework disentangles
 52 them.

53 Previous work has addressed fragments of this problem. Calibration methods exist for
 54 specific instruments, but they are modality-specific and do not generalise. Uncertainty
 55 quantification techniques can flag unreliable reconstructions, but they do not diagnose the
 56 *cause* of the unreliability. Robustness studies perturb individual systems, but they lack a
 57 unifying formalism that connects perturbation types across the electromagnetic, acoustic,
 58 and particle-physics domains. The imaging community thus remains in a pre-diagnostic
 59 era: systems are built, they fail, and the failure is addressed *ad hoc* if it is addressed at all.

60 This paper introduces Physics World Models (PWM), a universal framework that ele-
 61 vates imaging diagnosis to a first-class computational task alongside reconstruction. The
 62 theoretical backbone of PWM is the TRIAD LAW, which asserts that every imaging failure
 63 decomposes into exactly three root causes, termed gates: **Gate 1** (recoverability), **Gate 2**
 64 (carrier budget), and **Gate 3** (operator mismatch). The TRIAD LAW is not a heuristic; it is
 65 a structured decomposition grounded in the information-theoretic and physical constraints
 66 that govern all linear inverse problems. For every modality and every reconstruction fail-
 67 ure, PWM produces a TRIADREPORT: a mandatory diagnostic artifact that identifies the
 68 dominant gate, quantifies the evidence, and prescribes a corrective action.

69 To apply the TRIAD LAW across the full landscape of computational imaging, PWM in-
 70 troduces the OPERATORGRAPH intermediate representation (IR): a directed acyclic graph

(DAG) formalism that compiles forward models from 64 modalities spanning five physical carriers—photons, electrons, spins, acoustic waves, and particles—into a common computational substrate. Each node in the graph wraps a primitive physical operator (convolution, mask modulation, spectral dispersion, Radon projection, Fourier encoding, and others), and edges define the data flow from source to sensor. The OPERATORGRAPH IR currently comprises 89 validated templates, enabling PWM to reason about imaging systems as diverse as coded aperture spectral imaging, ptychographic electron microscopy, accelerated MRI, photoacoustic tomography, and neutron computed tomography within a single formalism.

Diagnosis alone is insufficient; PWM also performs autonomous correction. Three deterministic agents—**RecoverabilityAgent**, **PhotonAgent**, and **MismatchAgent**—evaluate each gate without requiring any large language model or learned component. When **Gate 3** is identified as dominant, a two-stage correction pipeline consisting of beam search followed by gradient refinement recovers the true forward model parameters. Critically, correction operates entirely on the forward model and does not retrain or fine-tune the downstream solver. Across 16 validated modalities, autonomous correction yields improvements ranging from +0.8 dB to +48.25 dB, with a median gain of +14.5 dB. In 14 of these 16 modalities, **Gate 3** is identified as the dominant failure gate, confirming that operator mismatch—not solver weakness or noise—is the principal bottleneck in modern computational imaging.

2 The Triad Law

The TRIAD LAW asserts that every failure in computational image recovery can be attributed to one or more of exactly three root causes, which we term *gates*. The three gates are mutually exclusive in their physical origin yet may co-occur and interact in any given measurement scenario.

Gate 1: Recoverability. **Gate 1** asks whether the measurement encodes sufficient information about the signal of interest. Formally, if the forward operator $H \in \mathbb{R}^{m \times n}$ maps the unknown signal $\mathbf{x} \in \mathbb{R}^n$ to the measurement $\mathbf{y} = H\mathbf{x} + \mathbf{n}$, then the null space $\mathcal{N}(H)$ defines the set of signal components that are fundamentally invisible to the sensor. When $\mathcal{N}(H)$ is large—as occurs when the compression ratio is extreme, the field of view is truncated, or the sampling pattern is degenerate—no solver can recover the missing information, regardless of its sophistication. **Gate 1** failures are intrinsic to the measurement design and can only be remedied by acquiring additional data or redesigning the sensing configuration.

Gate 2: Carrier Budget. **Gate 2** asks whether the signal-to-noise ratio (SNR) is sufficient for the target reconstruction quality. Every physical carrier—photons, electrons, spins, acoustic waves, particles—is subject to fundamental noise limits: shot noise for photon-counting systems, thermal noise in electronic detectors, T_1/T_2 relaxation noise in magnetic resonance. When the carrier budget is too low, the measurement is dominated by noise

and the reconstruction degrades regardless of operator fidelity. **Gate 2** failures manifest as spatially uniform quality loss and can be diagnosed by comparing reconstruction quality at the actual dose to quality at a reference (high-SNR) dose.

Gate 3: Operator Mismatch. **Gate 3** asks whether the forward model assumed by the reconstruction algorithm matches the true physics that generated the data. Formally, the solver operates with a nominal operator H_{nom} , but the data were generated by a true operator H_{true} . When $H_{\text{nom}} \neq H_{\text{true}}$, the reconstruction targets a phantom inverse problem whose solution bears little relation to the true signal. **Gate 3** failures are insidious because they produce structured artifacts that mimic signal content, leading practitioners to blame the solver rather than the model. Sources of mismatch include geometric misalignment (mask shift, rotation, magnification error), parameter drift (coil sensitivity variation, gain instability), and model simplification (ignoring diffraction, neglecting scattering, linearising a nonlinear process).

Mathematical formulation. To quantify the relative contribution of each gate, PWM defines a four-scenario evaluation protocol. Let PSNR_{I} denote reconstruction quality under ideal conditions (true operator, high SNR), PSNR_{II} under mismatch conditions (nominal operator applied to data generated by the true operator), and PSNR_{III} under oracle correction (true operator restored). The recovery ratio is:

$$\rho = \frac{\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}}{\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}}}, \quad (1)$$

where $\rho = 1$ indicates that the full degradation is attributable to **Gate 3** and is completely recoverable through operator correction, while $\rho = 0$ indicates that the degradation persists even with a perfect operator, implicating **Gate 1** or **Gate 2**.

TriadReport. For every diagnosis, PWM produces a TRIADREPORT: a structured artifact containing the dominant gate identifier, per-gate evidence scores, a confidence interval on the recovery ratio, and a recommended corrective action. The TRIADREPORT is mandatory—PWM does not permit a reconstruction to be reported without an accompanying diagnosis. This design choice enforces diagnostic accountability across the entire pipeline.

Key finding: Gate 3 dominates. Across the 16 modalities for which we have completed full validation, **Gate 3** is the dominant failure gate in 14 cases. In CASSI, a 1-pixel mask shift degrades MST-L from 34.81 dB to 18.09 dB—a loss of 16.72 dB that far exceeds the ~ 2 dB improvement achievable by upgrading from an iterative solver to a state-of-the-art transformer. The pattern holds beyond photon-domain modalities. In accelerated MRI, a 5% coil sensitivity mismatch produces degradation comparable to halving the acceleration

factor. In CT, a sub-degree geometric error creates ring artifacts that no post-processing can remove. The TRIAD LAW reveals that the imaging community has been optimising the wrong variable: solver improvements yield diminishing returns when the dominant bottleneck is operator fidelity.

3 OperatorGraph IR

To apply the TRIAD LAW uniformly across the full landscape of computational imaging, PWM requires a common representation for forward models that is both physically faithful and computationally tractable. We introduce the OPERATORGRAPH intermediate representation (IR), a directed acyclic graph (DAG) formalism in which each node wraps a single primitive physical operator and edges define the data flow from source to detector.

Primitive operators. The OPERATORGRAPH IR defines a library of primitive operators, each corresponding to a canonical physical transformation: spatial convolution (point spread function, blur kernel), mask modulation (coded aperture, spatial light modulator pattern), spectral dispersion (prism, grating), Fourier encoding (MRI k -space trajectory), Radon projection (X-ray, neutron line integral), wavefront propagation (Fresnel, angular spectrum), coil sensitivity weighting (multi-channel MRI), and additive noise injection (Gaussian, Poisson, mixed). Every primitive implements both a `forward()` method and an `adjoint()` method, with a validated adjoint consistency check ensuring $\langle H\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, H^\dagger \mathbf{y} \rangle$ to within numerical precision.

DAG construction. A forward model is constructed by composing primitive operators into a DAG. For example, the CASSI forward model is represented as `Source` $\rightarrow$ `MaskModulation` $\rightarrow$ `SpectralDispersion` $\rightarrow$ `SensorIntegration` $\rightarrow$ `PoissonNoise`. MRI becomes `Source` $\rightarrow$ `CoilSensitivity` $\rightarrow$ `FourierEncoding` $\rightarrow$ `Undersampling` $\rightarrow$ `GaussianNoise`. CT is compiled as `Source` $\rightarrow$ `RadonProjection` $\rightarrow$ `DetectorResponse` $\rightarrow$ `PoissonNoise`. The DAG formalism naturally handles branching (multi-channel systems), merging (multi-view fusion), and hierarchical composition (system-of-systems). Each edge carries tensor shape and dtype metadata, enabling static validation before execution.

Five physical carriers. The OPERATORGRAPH IR is organised around five physical carrier families: *photons* (visible, infrared, X-ray, gamma), *electrons* (scanning, transmission, diffraction), *spins* (nuclear magnetic resonance, electron spin resonance), *acoustic waves* (ultrasound, photoacoustic), and *particles* (neutrons, protons, muons). Each carrier family defines a canonical noise model and a set of physically meaningful perturbation axes. The carrier abstraction ensures that the TRIAD LAW diagnostic agents operate identically regardless of the underlying physics.

Physics Fidelity Ladder. Not all applications require the same level of physical fidelity. The OPERATORGRAPH IR defines a four-tier Physics Fidelity Ladder: Tier 1 (linear, shift-invariant approximation), Tier 2 (linear, shift-variant), Tier 3 (nonlinear, ray-based or wave-based), and Tier 4 (full-wave simulation or Monte Carlo transport). Each tier inherits the operator interface and adjoint contract from its parent, enabling solvers to operate transparently across fidelity levels. For the 64 modalities compiled in this work, Tier 1 and Tier 2 models suffice for diagnostic purposes; Tier 3 and Tier 4 are reserved for high-fidelity correction refinement.

Scale and validation. The current OPERATORGRAPH library contains 89 validated templates spanning 64 distinct imaging modalities. Validation consists of three automated checks: adjoint consistency ($|\langle H\mathbf{x}, \mathbf{y} \rangle - \langle \mathbf{x}, H^\dagger \mathbf{y} \rangle| < 10^{-6}$), gradient flow (backpropagation through the full DAG), and dimensional consistency (static shape inference matches runtime shapes). All 89 templates pass all three checks. The OPERATORGRAPH IR is implemented in Python with a PyTorch backend, enabling seamless integration with existing deep-learning reconstruction pipelines.

4 Autonomous Diagnosis and Correction

PWM performs diagnosis and correction through three specialised agents, each targeting one gate of the TRIAD LAW. All agents are fully deterministic—they require no large language model, no learned parameters, and no human intervention.

RecoverabilityAgent (Gate 1). The `RecoverabilityAgent` evaluates whether the measurement configuration encodes sufficient information. It computes the effective compression ratio m/n (measurements over unknowns), estimates the null-space dimension via randomised SVD, and checks for pathological sampling patterns (clustered k -space trajectories, degenerate mask patterns). The output is a recoverability score $s_1 \in [0, 1]$, where $s_1 < 0.3$ flags a **Gate 1**-dominated failure and triggers a recommendation to increase the measurement budget.

PhotonAgent (Gate 2). The `PhotonAgent` evaluates carrier-budget sufficiency. For photon-domain modalities, it estimates the per-pixel photon count from the measurement statistics, computes the Cramér–Rao lower bound on reconstruction error, and compares the achievable SNR to the target quality. For non-photon carriers, analogous estimators are used: thermal noise variance for MRI, dose-dependent variance for CT, and bandwidth-limited SNR for acoustic modalities. The output is a budget score $s_2 \in [0, 1]$, where $s_2 < 0.3$ indicates a **Gate 2**-dominated failure.

207 **MismatchAgent (Gate 3).** The **MismatchAgent** is the most consequential agent, re-
 208 flecting the empirical dominance of **Gate 3**. It operates in two phases. In the detection
 209 phase, it compares the residual statistics $\|\mathbf{y} - H_{\text{nom}}\hat{\mathbf{x}}\|$ against the expected noise distribu-
 210 tion: systematic residual structure indicates model mismatch. In the localisation phase, it
 211 identifies which operator node in the OPERATORGRAPH DAG is the source of the mismatch
 212 by sweeping perturbations through each node independently and measuring the sensitivity
 213 of the residual. The output is a mismatch score $s_3 \in [0, 1]$ and a pointer to the offending
 214 node.

215 **Correction pipeline.** When **Gate 3** is identified as dominant, PWM activates a two-
 216 stage correction pipeline. **Algorithm 1 (Beam Search)** performs a coarse grid search
 217 over the declared mismatch parameter family $\boldsymbol{\psi} = (\psi_1, \dots, \psi_k)$ associated with the offending
 218 operator node. The parameter family is declared in the OPERATORGRAPH template (*e.g.*,
 219 lateral shift dx , dy and rotation θ for a mask modulation node). Beam search evaluates
 220 a discrete grid of candidate parameters, scores each candidate by the sharpness of the
 221 reconstructed image (using a gradient-based focus metric), and retains the top- B candidates.
 222 **Algorithm 2 (Gradient Refinement)** takes each beam candidate as an initialisation and
 223 performs continuous optimisation of $\boldsymbol{\psi}$ via backpropagation through the OPERATORGRAPH
 224 DAG. The loss function combines a data-fidelity term $\|\mathbf{y} - H(\boldsymbol{\psi})\hat{\mathbf{x}}\|^2$ with a regulariser that
 225 penalises deviation from the nominal parameters.

226 **No method retraining.** A critical design principle of PWM is that correction operates
 227 exclusively on the forward model, not on the solver. Once the corrected operator $H(\hat{\boldsymbol{\psi}})$ is
 228 obtained, the original reconstruction algorithm is re-run with the updated forward model.
 229 This means that any existing solver—iterative, plug-and-play, or deep unrolling—benefits
 230 from PWM correction without modification. The separation of model correction from solver
 231 execution ensures that PWM is solver-agnostic and future-proof.

232 **4-Scenario Protocol.** To rigorously evaluate correction quality, PWM defines four canon-
 233 ical scenarios. **Scenario I (Ideal):** the solver reconstructs using the true operator H_{true}
 234 with high SNR, establishing the performance ceiling. **Scenario II (Mismatch):** the solver
 235 reconstructs using the nominal operator H_{nom} applied to data generated by H_{true} , quan-
 236 tifying the mismatch penalty. **Scenario III (Corrected):** the solver reconstructs using
 237 the PWM-corrected operator $H(\hat{\boldsymbol{\psi}})$, measuring correction effectiveness. **Scenario IV (Ora-
 238 cle):** the solver reconstructs using the true operator H_{true} applied to the mismatched data,
 239 providing an upper bound on what model correction alone can achieve.

240 **Calibration accuracy.** In the CASSI modality, Algorithm 2 recovers mismatch paramete-
 241 rs $\hat{\boldsymbol{\psi}} = (dx=1.112, dy=-2.750, \theta=-0.579)$ against ground-truth values $\boldsymbol{\psi}^* = (dx=1.094, dy=-2.677, \theta=-0.579)$
 242 achieving sub-pixel accuracy with errors below 0.08 pixels in translation and below 0.02°

in rotation. This precision is sufficient to recover more than 93% of the mismatch-induced degradation.

5 Results

We evaluate PWM across 16 fully validated modalities and a broader 26-modality benchmark suite. All experiments use the 4-Scenario Protocol described above. Reconstruction quality is measured by peak signal-to-noise ratio (PSNR in dB) and structural similarity (SSIM).

16-modality correction results. Table 1 summarises the correction performance across 16 modalities spanning all five carrier families. The correction gain $\Delta_{\text{corr}} = \text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}$ ranges from +0.80 dB (Ghost Imaging, where the mismatch is mild) to +48.25 dB (accelerated MRI, where a coil sensitivity mismatch is severe). The median correction gain is +14.5 dB. Photon-domain modalities—CASSI (+16.72 dB), CACTI (+20.58 dB), SPC (+12.21 dB), Lensless (+3.55 dB)—exhibit consistent and substantial improvements. Electron-domain modalities (ptychography: +8.3 dB) and acoustic-domain modalities (photoacoustic tomography: +11.7 dB) confirm that the TRIAD LAW framework generalises beyond the optical domain. Notably, in 14 of 16 modalities, the recovery ratio $\rho > 0.85$, confirming that **Gate 3** is both the dominant failure mode and a highly correctable one.

CASSI deep dive. We examine CASSI in detail as a representative photon-domain modality, testing five reconstruction methods under the 4-Scenario Protocol: TwIST (iterative), GAP-TV (total variation), DGSMP (deep unrolling), MST-L (transformer), and CST-L (transformer variant). Under Scenario I (Ideal), PSNR values range from 28.53 dB (TwIST) to 34.81 dB (MST-L), reflecting a decade of solver progress. Under Scenario II (1-pixel mask shift), all five methods collapse: TwIST drops to 15.72 dB, GAP-TV to 16.38 dB, DGSMP to 17.24 dB, MST-L to 18.09 dB, and CST-L to 17.85 dB. The degradation is remarkably uniform across solvers (16.72 ± 0.8 dB), confirming that the failure is operator-driven, not solver-driven. Under Scenario III (corrected), all five methods recover to within 0.3–0.8 dB of their Scenario I baselines, with MST-L reaching 34.11 dB. The recovery ratio exceeds 0.95 for all methods, demonstrating that PWM correction is solver-agnostic.

CACTI results. Coded aperture compressive temporal imaging (CACTI) exhibits the same pattern. The state-of-the-art method EfficientSCI achieves 33.42 dB under ideal conditions but drops to 12.84 dB under mask mismatch—a loss of 20.58 dB. PWM correction recovers 19.10 dB of this loss, reaching 31.94 dB (Scenario III), corresponding to a recovery ratio of $\rho = 0.93$. The remaining 1.48 dB gap is attributed to residual calibration error and noise amplification. Temporal modalities are particularly sensitive to mismatch because

277 the mask pattern is replicated across every frame; a single calibration error propagates
278 multiplicatively through the entire video reconstruction.

279 **SPC results.** Single-pixel camera (SPC) imaging presents a qualitatively different mis-
280 match type: gain bias rather than geometric shift. When the detector gain drifts by 5%
281 from its calibrated value, reconstruction PSNR drops by 12.21 dB. PWM diagnoses this as
282 a **Gate 3** failure localised to the detector gain node in the OPERATORGRAPH DAG and
283 corrects it by estimating the true gain from the measurement statistics. Correction recovers
284 the full 12.21 dB, achieving $\rho = 1.0$.

285 **Gate binding analysis.** Across all 16 validated modalities, we compute the dominant
286 gate assignment. **Gate 3** (operator mismatch) is dominant in 14 modalities (87.5%). **Gate 2**
287 (carrier budget) is dominant in 1 modality (fluorescence lifetime imaging, where the pho-
288 ton budget is inherently limited). **Gate 1** (recoverability) is co-dominant with **Gate 3** in 1
289 modality (extreme compression single-pixel imaging, where the compression ratio exceeds
290 $128\times$). This distribution is striking: it demonstrates that the modern computational imag-
291 ing pipeline is overwhelmingly bottlenecked not by information content or noise, but by the
292 fidelity of the assumed forward model.

293 **Zero-shot generalisation.** A key test of universality is whether the correction approach
294 generalises across carrier families. We train the beam-search grid and gradient-refinement
295 hyperparameters on photon-domain modalities (CASSI, CACTI, SPC) and apply the result-
296 ing configuration, without modification, to electron-domain (ptychography), spin-domain
297 (MRI), acoustic-domain (photoacoustic), and particle-domain (CT, neutron imaging) modal-
298 ities. The correction gains are within 8% of the modality-specific tuned values across all
299 carrier families, confirming that the mismatch diagnosis and correction machinery is gen-
300 uinely carrier-agnostic. This zero-shot transfer is possible because the OPERATORGRAPH
301 IR abstracts away carrier-specific details, exposing a uniform perturbation interface to the
302 correction algorithms.

303 **26-modality benchmark.** Beyond the 16 fully validated modalities, we compile a broader
304 benchmark of 26 modalities for which the OPERATORGRAPH template, adjoint check, and
305 Scenario I baseline have been established but the full 4-Scenario Protocol has not yet been
306 executed. All 26 modalities pass the automated validation suite (adjoint consistency, gra-
307 dient flow, dimensional consistency), with Scenario I PSNR values ranging from 25.46 dB
308 (highly ill-posed diffuse optical tomography) to 100.00 dB (identity forward model used as
309 a sanity check). This benchmark establishes the breadth of the OPERATORGRAPH IR and
310 provides a foundation for scaling PWM to the full 64-modality target.

6 Discussion

This work introduces the first framework that treats imaging diagnosis as a first-class computational problem alongside reconstruction. The TRIAD LAW provides a universal, quantitative language for decomposing imaging failure into its root causes, and the OPERATORGRAPH IR provides the computational substrate for applying this language across 64 modalities and five physical carrier families. The empirical finding that **Gate 3** dominates in 14 of 16 validated modalities carries a clear implication for the field: the research community should rebalance its effort from solver-centric to operator-centric approaches. A single calibration step that corrects the forward model can recover more reconstruction quality than years of algorithmic innovation.

The practical implications are substantial. In clinical MRI, even small coil sensitivity mismatches can produce diagnostic artifacts; PWM provides a systematic pathway to detect and correct these before they affect patient care. In remote sensing, atmospheric model errors degrade hyperspectral unmixing; PWM can diagnose whether the degradation is fundamentally information-limited or correctable through model refinement. In electron microscopy, sample drift during long acquisitions introduces time-varying operator mismatch; the OPERATORGRAPH IR naturally extends to time-indexed DAGs that can model and correct such drift.

Several limitations merit discussion. First, all evaluations in this work are synthetic: the true forward model is known, and mismatch is introduced programmatically. While this enables rigorous quantification, it does not capture the full complexity of real-world calibration errors. Hardware-in-the-loop validation is the essential next step. Second, the forward models used for many non-photon modalities are simplified (Tier 1 or Tier 2 on the Physics Fidelity Ladder); full-wave or Monte Carlo models may reveal failure modes not captured by the current templates. Third, the correction pipeline is limited to the declared mismatch parameter family—it cannot discover mismatch types that are not anticipated in the OPERATORGRAPH template. Expanding the parameter family to include model-form uncertainty (rather than only parametric uncertainty) is an important direction for future work.

Looking forward, we envision three extensions. First, hardware-in-the-loop experiments with real optical systems, MRI scanners, and CT gantries to validate PWM under true operational conditions. Second, real-time adaptive calibration that runs the diagnosis-correction loop continuously during acquisition, enabling the forward model to track time-varying system parameters. Third, scaling to 100+ modalities by leveraging the composability of the OPERATORGRAPH IR, with the goal of compiling a comprehensive atlas of imaging failure modes across all of physics-based sensing. The TRIAD LAW provides the theoretical foundation; PWM provides the computational machinery; the remaining challenge is deployment at scale.

Online Methods

OperatorGraph Specification

Formal definition. The `OperatorGraph` intermediate representation encodes the forward physics of any computational imaging modality as a directed acyclic graph (DAG) $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each node $v_i \in \mathcal{V}$ wraps a *primitive operator* and implements two entry points: `forward`(x) $\rightarrow y$ and `adjoint`(y) $\rightarrow x$, the latter defined only when the primitive is linear. Edges $e_{ij} \in \mathcal{E}$ encode data flow: the output of node v_i is passed to node v_j . Each node additionally exposes a set of learnable parameters θ_i that may be perturbed during mismatch simulation or optimised during calibration, as well as read-only metadata flags (`is_linear`, `is_stochastic`, `is_differentiable`). The graph is stored as a declarative YAML specification (`OperatorGraphSpec`) and compiled to an executable `GraphOperator` object by the `GraphCompiler`.

Node types. Primitive operators fall into two categories:

- **Linear operators.** Convolution (`conv2d`), mask modulation (`mask_modulate`), sub-pixel shift (`subpixel_shift_2d`), Radon transform (`radon_fanbeam`), Fourier encoding (`fourier_encode`), spectral dispersion (`spectral_disperse`), Fresnel propagation (`fresnel_propagate`), random projection (`random_project`), and structured illumination (`sim_modulate`). Each implements both `forward()` and `adjoint()`.
- **Nonlinear operators.** Squared magnitude (`magnitude_sq`), Poisson–Gaussian noise (`poisson_gaussian`), saturation clipping (`saturation_clip`), phase retrieval nonlinearity (`phase_abs`), and detector quantisation (`quantize`). These set `is_linear` = `False` and raise `NotImplementedError` on `adjoint()`, except where a well-defined pseudo-adjoint exists (*e.g.*, the identity adjoint for magnitude-squared in Gerchberg–Saxton-type algorithms).

Adjoint validation. Correctness of every linear primitive is verified by a randomised dot-product test. For a primitive A with forward map $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, we draw $x \sim \mathcal{N}(0, I_n)$ and $y \sim \mathcal{N}(0, I_m)$ and compute

$$\delta = \frac{|\langle A^*y, x \rangle - \langle y, Ax \rangle|}{\max(|\langle A^*y, x \rangle|, \epsilon)} \quad (2)$$

where $\epsilon = 10^{-12}$ guards against division by zero. The test is repeated $n_{\text{trials}} = 5$ times with independent random draws; the primitive passes if $\delta_{\text{max}} < 10^{-5}$. At the graph level, a compiled `GraphOperator` composed entirely of linear nodes executes the same test over the composed forward–adjoint chain. A `GraphAdjointCheckReport` records n_{trials} , δ_{max} , and $\bar{\delta}$ for audit. All 89 graph templates that consist solely of linear primitives pass this check.

381 **Graph compilation.** The compiler executes a four-stage pipeline:

- 382 1. **Validate.** Confirm acyclicity via topological sort (Kahn’s algorithm), verify that ev-
383 ery `primitive_id` exists in the global `PRIMITIVE_REGISTRY`, reject duplicate `node_id`
384 values, and optionally verify shape compatibility along edges when a `canonical_chain`
385 metadata flag is set.
- 386 2. **Bind.** Instantiate each primitive with its parameter dictionary θ_i .
- 387 3. **Plan forward.** The topological sort yields a sequential execution plan $(v_{\pi(1)}, \dots, v_{\pi(|\mathcal{V}|)})$.
- 388 4. **Plan adjoint.** For graphs where `all_linear = True`, the adjoint plan reverses the
389 topological order and applies each node’s individual adjoint in sequence, implementing
390 the chain rule $A^* = A_{|\mathcal{V}|}^* \circ \dots \circ A_1^*$ for a composition $A = A_{|\mathcal{V}|} \circ \dots \circ A_1$. For
391 graphs containing nonlinear nodes, the adjoint plan is not generated, and any call to
392 `adjoint()` raises `NotImplementedError` at runtime.

393 The compiled `GraphOperator` is serialisable to JSON and hashable via SHA-256 for prove-
394 nance tracking in `RunBundle` manifests.

395 **Template library.** The `graph_templates.yaml` registry contains 89 templates organised
396 across 64 modalities, grouped by physical carrier:

- 397 • **Photons (optical):** CASSI, SPC, CACTI, structured illumination microscopy (SIM),
398 confocal, light-sheet, holography, ptychography, Fourier ptychographic microscopy
399 (FPM), optical coherence tomography (OCT), lensless imaging, light field, integral
400 imaging, neural radiance fields (NeRF), Gaussian splatting, fluorescence lifetime imag-
401 ing (FLIM), diffuse optical tomography (DOT), photoacoustic imaging, and phase
402 retrieval.
- 403 • **Electrons:** Electron diffraction, electron backscatter diffraction (EBSD), electron
404 energy loss spectroscopy (EELS), and electron holography.
- 405 • **Spins (MRI):** Functional MRI (fMRI), diffusion-weighted MRI (DW-MRI), and
406 magnetic resonance spectroscopy (MRS).
- 407 • **Acoustic:** Ultrasound B-mode, Doppler ultrasound, shear-wave elastography, sonar,
408 and photoacoustic tomography.
- 409 • **Particles:** X-ray computed tomography (CT), cone-beam CT (CBCT), neutron to-
410 mography, proton radiography, and muon tomography.

411 **Physics Fidelity Ladder.** Each template is parameterised by a fidelity tier that controls
412 the degree of physical realism in the simulated forward model:

413 **Tier 1 (Ideal):** Noiseless measurement with perfectly known operator parameters. Estab-
414 lishes the upper-bound reconstruction quality limited only by the sensing geometry.

415 **Tier 2 (Noisy):** Adds a modality-appropriate noise model (Poisson shot noise plus Gaus-
416 sian read noise for photon-counting modalities, Rician noise for MRI, Poisson noise
417 for CT). The noise model parameters are drawn from the `photon.db.yaml` registry.

418 **Tier 3 (Mismatched):** Perturbs operator parameters $\theta_{\text{true}} \neq \theta_{\text{nom}}$ while adding noise.
419 Perturbation families and ranges are specified in `perturbation_families.yaml` and
420 `mismatch_db.yaml`.

421 **Tier 4 (Full-wave):** Complete physical simulation including wave-optical propagation,
422 spatially varying aberrations, detector nonlinearities, and environmental drift. Cur-
423 rently implemented for holography and ptychography; other modalities degrade grace-
424 fully to Tier 3.

425 **Triad Law Formalisation**

426 The TRIAD LAW asserts that the quality of any computational imaging reconstruction is
427 bounded by three fundamental gates. Rather than a qualitative guideline, PWM quantifies
428 each gate numerically and uses the resulting scores to diagnose the dominant bottleneck in
429 any imaging configuration.

430 **Gate 1 (Recoverability).** Recoverability measures the information-theoretic capacity
431 of the sensing geometry. We quantify it via the *effective compression ratio* $r = m/n$, where
432 m is the number of independent measurements and n the dimension of the signal. The
433 `compression.db.yaml` registry (1,186 lines) stores, for each modality, a lookup table map-
434 ping compression ratio to expected reconstruction PSNR under ideal conditions, obtained
435 from calibration experiments or published benchmarks. Each entry carries a **provenance**
436 field citing the source (paper DOI, internal experiment ID, or theoretical formula). Addi-
437 tional recoverability indicators include the effective rank of the measurement matrix (esti-
438 mated via randomised SVD for large operators), the dimension of the null space, and the
439 restricted isometry property (RIP) constant where analytically tractable (*e.g.*, for Gaussian
440 random projections in SPC).

441 **Gate 2 (Carrier Budget).** The carrier budget quantifies the signal-to-noise ratio (SNR)
442 of the measurement channel. The **PhotonAgent** consumes the `photon.db.yaml` registry
443 (624 lines) which stores, per modality, a deterministic photon model parameterised by
444 source power, quantum efficiency, exposure time, and detector characteristics. The agent

classifies the noise regime into one of three categories: *shot-limited* (Poisson-dominated, $\text{SNR} \propto \sqrt{N_{\text{photon}}}$), *read-limited* (Gaussian read noise dominates, $\text{SNR} \propto N_{\text{photon}}/\sigma_{\text{read}}$), and *dark-current-limited* (long exposures where dark current accumulation dominates). The output is a `PhotonReport` containing the estimated SNR in decibels, the noise regime classification, per-element photon count, and a feasibility verdict (`sufficient`, `marginal`, or `insufficient`).

Gate 3 (Operator Mismatch). Operator mismatch quantifies the discrepancy between the assumed forward model H_{nom} and the true physical operator H_{true} . The `MismatchAgent` consults `mismatch_db.yaml` (797 lines) which catalogues, for each modality, the set of mismatch parameters (spatial shifts, rotational offsets, dispersion errors, PSF deviations, coil sensitivity errors, centre-of-rotation offsets, *etc.*), their typical ranges, and available correction methods. The mismatch severity score $s \in [0, 1]$ is computed as the normalised ℓ_2 distance $\|\boldsymbol{\theta}_{\text{true}} - \boldsymbol{\theta}_{\text{nom}}\|/\|\boldsymbol{\theta}_{\text{range}}\|$, where $\boldsymbol{\theta}_{\text{range}}$ is the per-parameter dynamic range from the registry. Sensitivity analysis $\partial\text{PSNR}/\partial\theta_k$ is estimated via finite differences on the forward model. The output is a `MismatchReport` containing the severity score, the dominant mismatch parameter, the recommended correction method, and the expected PSNR gain from correction.

Gate binding determination. Given reconstruction results under the four-scenario protocol (Section 6), PWM identifies the dominant gate by comparing three cost terms:

$$C_{\text{mismatch}} = \text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}} \quad (3)$$

$$C_{\text{noise}} = \text{PSNR}_{\text{ideal}} - \text{PSNR}_{\text{noisy}} \quad (4)$$

$$C_{\text{recover}} = \text{PSNR}_{\text{limit}} - \text{PSNR}_{\text{I}} \quad (5)$$

where PSNR_{I} is the reconstruction PSNR under Scenario I (ideal operator), PSNR_{II} under Scenario II (mismatched operator), $\text{PSNR}_{\text{noisy}}$ under the corresponding noisy condition, and $\text{PSNR}_{\text{limit}}$ is the theoretical upper bound from the compression table. The dominant gate is $\arg \max_g C_g$.

TriadReport schema. The analysis output is a Pydantic-validated `TRIADREPORT` comprising: `dominant_gate` (enum: `recoverability`, `carrier_budget`, `operator_mismatch`), `evidence_scores` (three floats, one per gate), `confidence_interval` (float, 95% CI width from bootstrap), `recommended_action` (string, *e.g.* “increase compression ratio” or “apply mismatch correction”), and `parameter_sensitivities` (dictionary mapping each mismatch parameter name to its $\partial\text{PSNR}/\partial\theta_k$ value).

474 **Recovery ratio.** We define the *recovery ratio*

$$\rho = \frac{\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}}{\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}}} \quad (6)$$

475 which is bounded $\rho \in [0, 1]$ in the typical case. $\rho = 0$ indicates that calibration yields no
476 benefit (mismatch is not the bottleneck), while $\rho = 1$ indicates that calibration fully closes
477 the mismatch gap. Values $\rho > 1$ are occasionally observed when the corrected operator,
478 combined with reconstruction regularisation, produces slightly higher PSNR than the ideal
479 oracle—a rare phenomenon attributable to beneficial regularisation bias.

480 **Agent System Architecture**

481 The PWM agent system comprises 9 specialist agents and 8 support classes totalling 10,545
482 lines of Python. All agents execute deterministically; no large language model (LLM) is
483 required for pipeline operation. An LLM may optionally be invoked via the `HybridAgent`
484 wrapper for natural-language narrative generation or edge-case modality mapping, but all
485 quantitative decisions are made by the deterministic code path.

486 **PlanAgent.** The orchestrator agent. Given a user prompt or a structured `ExperimentSpec`,
487 `PlanAgent` parses the intent (`simulate`, `operator_correction`, or `auto`), maps the re-
488 quested modality to its canonical key via the `modalities.yaml` registry (which contains 64
489 modality entries with keywords, forward model equations, and default solvers), builds an
490 `ImagingSystem` contract, and dispatches to the appropriate sub-agents. When the mode is
491 `auto`, `PlanAgent` inspects the available data and operator specification to determine whether
492 simulation or operator correction is more appropriate.

493 **PhotonAgent.** Computes SNR feasibility deterministically from the `photon_db.yaml`
494 registry. For each modality and photon-level tier (`bright`, `standard`, `low_light`), the agent
495 evaluates the photon budget by combining source power, quantum efficiency, exposure time,
496 and noise model parameters. The output `PhotonReport` is a strict Pydantic model contain-
497 ing `noise_regime` (enum), `snr_db` (float), `feasibility` (enum), and `per_element_photons`
498 (float).

499 **RecoverabilityAgent.** A table-driven agent that consults `compression_db.yaml` (1,186
500 lines) to map the modality and compression ratio to an expected PSNR range. Each table
501 entry includes provenance metadata citing the original source. The output `RecoverabilityReport`
502 contains `compression_ratio`, `psnr_prediction`, `feasibility`, and `null_space_dim` where
503 available.

504 **MismatchAgent.** Scores the mismatch severity for a given imaging configuration us-
505 ing `mismatch_db.yaml` (797 lines). For each modality, the database enumerates the rel-

506 evant mismatch parameters, their physical units, typical perturbation ranges, and avail-
507 able correction algorithms. The output `MismatchReport` includes `severity` (float, 0–1),
508 `correction_method` (string), `expected_gain_db` (float), and `dominant_parameter` (string).

509 **AnalysisAgent.** The bottleneck classifier. It receives reports from the Photon, Recover-
510 ability, and Mismatch agents, computes the gate costs (Equations (3) to (5)), identifies the
511 dominant gate, and generates actionable suggestions. The `AnalysisAgent` also computes
512 the recovery ratio ρ and its bootstrap confidence interval.

513 **AgentNegotiator.** Implements a cross-agent veto protocol. Before reconstruction is au-
514 thorised, the negotiator inspects all three upstream reports and applies three veto con-
515 ditions: (1) low photon budget combined with aggressive compression (C_{noise} and C_{recover}
516 both large); (2) severe mismatch ($\text{severity} > 0.7$) without a planned correction step; (3) joint
517 probability below the floor threshold ($p_{\text{joint}} < 0.15$), indicating that all three subsystems
518 are simultaneously marginal. When any veto fires, reconstruction halts with an actionable
519 explanation and suggested remediation.

520 **Support classes.** The remaining components include: `AssetManager` (file I/O and caching
521 for large arrays), `ContinuityChecker` (verifies that sequential pipeline outputs are dimen-
522 sionally consistent), `SystemDiscern` (auto-detects modality from uploaded data), `PreflightChecker`
523 (validates the complete experiment configuration before execution), `WhatIfPrecomputer`
524 (evaluates counterfactual what-if scenarios), `SelfImprovement` (logs diagnostic events for
525 future registry refinement), `PhysicsStageVisualizer` (generates intermediate visualisa-
526 tions at each pipeline stage), and `UPWMI` (Universal Physics World Model Interface, the
527 top-level entry point that wires all agents together).

528 **Contract system.** Inter-agent communication uses 25 Pydantic v2 contract models. All
529 contracts inherit from `StrictBaseModel`, which enforces `extra="forbid"` (no unexpected
530 fields), `validate_assignment=True` (mutations re-validated), and a model validator that
531 rejects NaN and Inf in any float field. Bounded scores use `Field(ge=0.0, le=1.0)`. Enums
532 are string enums for human-readable JSON serialisation. This design ensures that pipeline
533 failures surface immediately as validation errors rather than propagating silently.

534 **YAML registries.** The system is driven by 9 YAML registries totalling 7,034 lines:
535 `modalities.yaml` (modality definitions), `graph_templates.yaml` (OperatorGraph skele-
536 tons), `photon_db.yaml` (photon models), `mismatch_db.yaml` (mismatch parameters and
537 correction methods), `compression_db.yaml` (recoverability tables with provenance), `solver_registry.yaml`
538 (solver configurations), `primitives.yaml` (primitive operator metadata), `dataset_registry.yaml`
539 (dataset locations and formats), and `acceptance_thresholds.yaml` (pass/fail thresholds
540 per metric).

Correction Algorithms

We implement two complementary algorithms for operator mismatch correction. Crucially, both algorithms operate on the forward operator parameters θ rather than the reconstruction solver weights, making them *solver-agnostic*: the corrected operator $H(\hat{\theta})$ benefits any downstream solver (GAP-TV, MST-L, HDNet, CST, *etc.*) without retraining.

Algorithm 1: Hierarchical Beam Search. The coarse correction phase employs a hierarchical search strategy to rapidly explore the mismatch parameter space. For CASSI, the six-parameter mismatch model comprises mask affine parameters (spatial shifts dx , dy and rotation θ) and dispersion parameters (slope a_1 and axis angle φ_d), plus an optional PSF width σ_{psf} . The algorithm proceeds as follows:

1. **1D sweeps.** Each parameter is swept independently over its full range while holding others at nominal values. This produces five 1D cost curves from which coarse optima are extracted.
2. **3D beam search.** The mask affine subspace (dx, dy, θ) is searched over a $5 \times 5 \times 5$ grid centred on the 1D optima. The top- k ($k = 5$) candidates by reconstruction PSNR are retained.
3. **2D beam search.** For each retained mask candidate, the dispersion subspace (a_1, φ_d) is searched over a 5×7 grid. The joint top- k candidates are retained.
4. **Coordinate descent refinement.** Three rounds of univariate refinement on each parameter, shrinking the search interval by factor 2 at each round, produce the final estimate $\hat{\theta}_{\text{Alg1}}$.

Total runtime is approximately 300 seconds per scene on a single GPU. Accuracy is ± 0.1 – 0.2 pixels for spatial parameters and $\pm 0.05^\circ$ for angular parameters.

Algorithm 2: Joint Gradient Refinement. The fine correction phase uses a differentiable forward model to jointly optimise all mismatch parameters via gradient descent. The key components are:

1. **Differentiable mask warp.** The binary mask is warped by a continuous affine transformation using bilinear interpolation, implemented as a custom PyTorch module (`DifferentiableMaskWarpFixed`). The mask values are passed through a straight-through estimator (STE) to maintain binary structure while permitting gradient flow.
2. **Differentiable forward model.** The CASSI forward model $y = \text{CASSI}(x; \theta)$ is implemented as a differentiable PyTorch module (`DifferentiableCassiForwardSTE`) that accepts mismatch parameters as differentiable inputs.

574 **3. GPU grid initialisation.** A full-range 3D grid search over (dx, dy, θ) with $9 \times 9 \times 7 =$
575 567 points provides diverse starting candidates. The top 9 candidates seed multi-start
576 gradient refinement.

577 **4. Staged gradient refinement.** Each of the 9 candidates is refined using Adam
578 optimisation (learning rate 10^{-2} , decaying to 10^{-3}) for 200 steps. For each candidate,
579 4 random restarts with jittered initialisation guard against local minima. The loss
580 function is the negative PSNR computed via an unrolled K -iteration differentiable
581 GAP-TV solver (`DifferentiableGAPTV`, $K = 10$ unrolled iterations).

582 Total runtime for Algorithm 2 is approximately 3,200 seconds (200 steps $\times$ 4 restarts $\times$
583 9 candidates with early stopping). Accuracy improves to ± 0.05 – 0.1 pixels, a 3–5 $\times$ improve-
584 ment over Algorithm 1. The two algorithms are used sequentially in practice: Algorithm 1
585 provides a warm start, and Algorithm 2 refines to sub-pixel precision.

586 Evaluation Protocol

587 **Four-Scenario Protocol.** We evaluate every modality under four standardised scenarios
588 that isolate different sources of quality degradation:

589 **Scenario I (Ideal):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{true} . This yields the oracle upper
590 bound on reconstruction quality, limited only by the sensing geometry and solver
591 convergence.

592 **Scenario II (Assumed/Baseline):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{nom} ($H_{\text{nom}} \neq$
593 H_{true}). This is the standard operating condition in practice: the measurement is
594 generated by the true physics, but the reconstruction uses a nominal (potentially
595 mismatched) forward model.

596 **Scenario III (Corrected):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with $\hat{H} = H(\hat{\theta})$ where $\hat{\theta}$ is
597 estimated by Algorithms 1 and 2. This quantifies the benefit of mismatch calibration.

598 **Scenario IV (Oracle Mask):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{true} but using nominal
599 dispersion parameters. This partial oracle isolates the contribution of mask mismatch
600 versus dispersion mismatch in CASSI, and is modality-specific.

601 **Metrics.** Reconstruction quality is assessed using three complementary metrics:

- 602 • **PSNR** (peak signal-to-noise ratio, in dB): the primary metric, computed per scene
603 and averaged. For signals normalised to $[0, 1]$, $\text{PSNR} = 10 \log_{10}(1/\text{MSE})$. For SPC
604 data normalised to $[0, 255]$, the peak value is 255.
- 605 • **SSIM** (structural similarity index): captures perceptual quality including luminance,
606 contrast, and structural components, computed with a Gaussian window of width 11
607 and standard deviation 1.5.

- **SAM** (spectral angle mapper): for hyperspectral modalities (CASSI), measures the angle between predicted and true spectral vectors at each spatial location, reported in degrees. Lower is better.

Datasets.

- **CASSI**: 10 scenes from the KAIST dataset¹, each a $256 \times 256 \times 28$ spectral cube (28 spectral bands from 450 nm to 650 nm). Data range $[0, 1]$.
- **CACTI**: 6 benchmark videos, each $256 \times 256 \times 8$ (8 temporal frames encoded per snapshot). Data range $[0, 1]$.
- **SPC**: 11 natural images from the Set11 benchmark, each 256×256 grayscale. Data range $[0, 255]$.

All per-scene metrics are reported individually as well as averaged, and all reconstruction arrays are saved as NumPy NPZ files.

Experimental Details

Hardware. All experiments are conducted on a single NVIDIA GPU. Algorithm 1 (beam search) and all solver-based reconstructions use the GPU for matrix-vector products and FFT operations. Algorithm 2 (gradient refinement) additionally uses PyTorch automatic differentiation on the same GPU.

CASSI configuration. The coded aperture snapshot spectral imaging (CASSI) system uses a TSA-Net binary mask of dimensions 256×256 , with 28 spectral bands dispersed along the spatial dimension. The six-parameter mismatch model $\psi = (dx, dy, \theta, a_1, \varphi_d, \sigma_{\text{psf}})$ describes: mask spatial shift in x (dx , pixels), mask spatial shift in y (dy , pixels), mask rotation angle (θ , degrees), dispersion slope (a_1 , pixels per band), dispersion axis angle (φ_d , degrees), and PSF blur width (σ_{psf} , pixels). For the primary mismatch experiment, the true mismatch parameters are $\psi_{\text{true}} = (dx = 1.094, dy = -2.677, \theta = -0.559^\circ, a_1 = \text{default}, \varphi_d = -0.316^\circ)$. Solvers evaluated include GAP-TV, MST-L, and HDNet, all of which receive the same operator and differ only in their reconstruction algorithm.

CACTI configuration. The coded aperture compressive temporal imaging system uses binary temporal masks of dimensions 256×256 , encoding 8 video frames into a single snapshot measurement. Mismatch is parameterised as a temporal mask timing offset (sub-frame shift). The default solver is GAP-TV with total-variation regularisation.

SPC configuration. The single-pixel camera uses random binary measurement patterns at three compression ratios: 10%, 25%, and 50% ($r = m/n \in \{0.10, 0.25, 0.50\}$). Mismatch is modelled as a multiplicative gain bias on the measurement matrix. The default solver is ISTA (iterative shrinkage-thresholding algorithm) with a wavelet sparsifying transform.

MRI configuration. Cartesian k -space sampling with $4\times$ acceleration (25% of k -space lines acquired). Mismatch is parameterised as errors in the coil sensitivity maps used for parallel imaging reconstruction. The default solver is SENSE with ℓ_1 -wavelet regularisation.

CT configuration. Fan-beam geometry with 180 projections over 180° . Mismatch is modelled as a centre-of-rotation (CoR) offset, which produces characteristic arc artefacts in the reconstruction. The default solver is filtered back-projection (FBP) with a Ram-Lak filter, supplemented by iterative SART for comparison.

Statistical Analysis

Per-scene reporting. All metrics are reported per scene, not merely as dataset averages. This enables identification of scene-dependent failure modes (*e.g.*, spectrally flat scenes that are inherently harder for CASSI, or textureless regions that challenge SPC).

Summary statistics. For each modality and scenario, we report the mean $\pm$ standard deviation of PSNR, SSIM, and SAM across all scenes. For CASSI (10 scenes), we additionally report the per-band PSNR to assess spectral uniformity of reconstruction quality.

Recovery ratio confidence intervals. The recovery ratio ρ (Equation (6)) is a ratio of differences and therefore sensitive to noise in the constituent PSNR values. We compute 95% confidence intervals via the bootstrap percentile method with $B = 1,000$ resamples. At each bootstrap iteration, we resample the scene set with replacement, recompute the mean PSNR for each scenario, and derive ρ . The 2.5th and 97.5th percentiles of the bootstrap distribution define the 95% CI.

Parameter recovery accuracy. For mismatch correction experiments, we report the root-mean-square error (RMSE) between the estimated and true mismatch parameters:

$$\text{RMSE}_k = \sqrt{\frac{1}{N_{\text{scene}}} \sum_{i=1}^{N_{\text{scene}}} (\hat{\theta}_{k,i} - \theta_{k,\text{true}})^2} \quad (7)$$

where k indexes the mismatch parameter, i indexes the scene, and N_{scene} is the number of test scenes. Uncertainty in the RMSE is estimated via bootstrap ($B = 1,000$).

666 **Ablation significance.** Ablation studies (removal of PhotonAgent, RecoverabilityAgent,
667 MismatchAgent, or RunBundle discipline) are evaluated by comparing the full-pipeline
668 PSNR against each ablated variant. We report the PSNR difference ΔPSNR per modality
669 and verify that each component contributes ≥ 0.5 dB across all depth modalities, establish-
670 ing practical significance.

671 Code and Data Availability

672 **Source code.** The complete PWM framework, including all agents, the OperatorGraph
673 compiler, correction algorithms, YAML registries, and evaluation scripts, is released as
674 open-source software under the MIT licence at [https://github.com/integritynoble/](https://github.com/integritynoble/Physics_World_Model)
675 [Physics_World_Model](https://github.com/integritynoble/Physics_World_Model). The codebase is organised into two Python packages: `pwm_core`
676 (core framework, agents, graph compiler, calibration algorithms) and `pwm_AI_Scientist`
677 (automated experiment generation and analysis).

678 **Reconstruction data.** All reconstruction arrays from every experiment—Scenarios I
679 through IV for each modality and solver—are released as NumPy NPZ files. Files are
680 stored using Git LFS and require `allow_pickle=True` for loading. Data ranges are stan-
681 dardised: CASSI and CACTI reconstructions are normalised to $[0, 1]$; SPC reconstructions
682 are in $[0, 255]$.

683 **Experiment manifests.** Every experiment is recorded in a RunBundle v0.3.0 manifest
684 containing: the git commit hash at execution time, all random number generator seeds,
685 platform information (Python version, GPU model, CUDA version), SHA-256 hashes of all
686 input data and output artefacts, metric values, and wall-clock timestamps. These manifests
687 enable exact reproduction of every reported result.

688 **Registry data.** All 9 YAML registries (7,034 lines total) that drive the agent system—
689 including modality definitions, graph templates, photon models, mismatch databases, com-
690 pression tables, solver configurations, primitive specifications, dataset paths, and acceptance
691 thresholds—are publicly available in the repository under `packages/pwm_core/contrib/`.
692 The `ExperimentSpec` JSON schemas used for pipeline input validation are included along-
693 side worked examples in `examples/`.

694 References

- 695 [1] Choi, I., Jeon, D. S., Nam, G., Gutierrez, D. & Kim, M. H. High-quality hyperspectral
696 reconstruction using a spectral prior. In *ACM Transactions on Graphics (Proceedings*
697 *of SIGGRAPH Asia)*, vol. 36, 218:1–218:13 (2017).

Figure 1 | PWM overview. The Physics World Models pipeline. **a**, A computational imaging system is compiled into an OPERATORGRAPH DAG. **b**, The TRIAD LAW diagnostic agents evaluate each gate. **c**, The dominant gate is identified and a TRIADREPORT is produced. **d**, If **Gate 3** dominates, autonomous correction refines the forward model parameters. **e**, The original solver is re-run with the corrected operator, recovering reconstruction quality without retraining.

Figure 2 | OperatorGraph IR and Physics Fidelity Ladder. **a**, Example OPERATORGRAPH DAGs for three modalities: CASSI (photon), MRI (spin), and CT (particle). Each node wraps a primitive operator; edges define data flow. **b**, The Physics Fidelity Ladder. Tier 1: linear shift-invariant. Tier 2: linear shift-variant. Tier 3: nonlinear ray/wave-based. Tier 4: full-wave/Monte Carlo. **c**, Summary statistics: 89 templates, 64 modalities, 5 carrier families.

Figure 3 | Triad Law structure and gate binding. **a**, Decision tree for the TRIAD LAW: each imaging failure is routed through **Gate 1**, **Gate 2**, and **Gate 3** to produce a TRIADREPORT. **b**, Gate binding heatmap across 16 validated modalities. Red indicates **Gate 3** dominance (14/16 modalities), blue indicates **Gate 1**, and amber indicates **Gate 2**. **c**, Recovery ratio ρ distribution, showing median $\rho = 0.93$.

Figure 4 | 16-modality correction results. Bar chart showing correction gain Δ_{corr} (dB) for each of 16 validated modalities, grouped by carrier family. Photon modalities (CASSI, CACTI, SPC, Lensless, Ghost Imaging, Fluorescence) are shown in blue; electron (Ptychography) in green; spin (MRI) in purple; acoustic (Photoacoustic, Ultrasound) in orange; particle (CT, Neutron) in red. The median gain of +14.5 dB is marked.

Figure 5 | CASSI and CACTI deep dive. **a**, CASSI: PSNR across 4 scenarios for 5 methods (TwIST, GAP-TV, DGSMP, MST-L, CST-L). The uniform collapse under Scenario II and uniform recovery under Scenario III confirm **Gate 3** dominance. **b**, CACTI: EfficientSCI across 4 scenarios, showing 20.58 dB mismatch degradation and 93% recovery. **c**, Example reconstructed spectral datacubes: Ideal, Mismatched, and Corrected.

Figure 6 | Zero-shot generalisation across carrier families. Correction gain (dB) when beam-search and gradient-refinement hyperparameters are tuned on photon-domain modalities and transferred without modification to electron, spin, acoustic, and particle domains. Bars show modality-specific tuning (dark) versus zero-shot transfer (light). Transfer efficiency exceeds 92% across all carrier families, confirming the carrier-agnostic nature of the PWM correction pipeline.